

Task 20: Git Branching, Merging, and Conflict Resolution

AIM

To implement branching strategies in Git by creating feature branches, performing merging and rebasing operations, and intentionally creating and resolving merge conflicts.

ALGORITHM

- Step 1: Create a feature branch from main using git checkout -b.
- Step 2: Make changes on the feature branch and commit.
- Step 3: Switch back to main and make conflicting changes to the same file.
- Step 4: Attempt to merge the feature branch into main to trigger a conflict.
- Step 5: Resolve the merge conflict manually by editing the file.
- Step 6: Complete the merge with a commit.
- Step 7: Demonstrate rebasing by creating another branch and using git rebase.
- Step 8: Verify clean history using git log --graph.

OUTPUT

Terminal Commands and Output:

Step 1: Create Feature Branch

```
$ git checkout -b feature-login  
Switched to a new branch 'feature-login'
```

Step 2: Make Changes on Feature Branch

```
$ echo "Login Page Content" > login.html  
$ echo "Welcome to the App" > index.html  
$ git add . && git commit -m "Add login page and update index"  
[feature-login b1c2d3e] Add login page and update index
```

Step 3: Make Conflicting Changes on Main

```
$ git checkout main  
$ echo "Home Page - Updated" > index.html  
$ git add . && git commit -m "Update index on main branch"
```

[main f4g5h6i] Update index on main branch

Step 4: Merge Feature Branch (Conflict!)

```
$ git merge feature-login
Auto-merging index.html
CONFLICT (content): Merge conflict in index.html
Automatic merge failed; fix conflicts and then commit.
```

Step 5: View and Resolve Conflict

```
$ cat index.html
<<<<<<< HEAD
Home Page - Updated
=====
Welcome to the App
>>>>>> feature-login

$ echo "Home Page - Updated | Welcome to the App" > index.html
```

Step 6: Complete Merge

```
$ git add index.html
$ git commit -m "Resolve merge conflict in index.html"
[main j7k8l9m] Resolve merge conflict in index.html
```

Step 7: Demonstrate Rebase

```
$ git checkout -b feature-dashboard
$ echo "Dashboard Content" > dashboard.html
$ git add . && git commit -m "Add dashboard"
$ git checkout main
$ echo "Footer added" >> index.html
$ git add . && git commit -m "Add footer to main"
$ git checkout feature-dashboard
$ git rebase main
Successfully rebased and updated refs/heads/feature-dashboard.
```

Step 8: View Graph History

```
$ git log --oneline --graph --all
* a1b2c3d (feature-dashboard) Add dashboard
* d4e5f6g (main) Add footer to main
* j7k8l9m Resolve merge conflict in index.html
| \
| * b1c2d3e Add login page and update index
| /
* f4g5h6i Update index on main branch
```

* a1b2c3d Initial commit

RESULT

Successfully demonstrated Git branching strategies including creating feature branches, merging with conflict resolution, and rebasing for a linear history. Merge conflicts were intentionally created and resolved manually.